PHARMACOGNOSY

Topic Outline

1. General Introduction

1.1 Definition

1.2 Brief history of Pharmacognosy

1.3 Scope of Pharmacognosy

Definition of pharmacognosy

- In a **broad sense**, *Pharmacognosy* embraces a knowledge of the history, distribution, cultivation, collection, selection, preparation, commerce, identification, evaluation, preservation, and use of drugs and economic substances affecting the health of man and other animals.

- Such economic substances extend beyond the category of crude drugs and their derivatives and include a variety of commercial products: allergens, allergenic extracts, antibiotics, immunizing biologicals, flavouring agents, condiments, beverages, insecticides, rodenticides, and herbicides. **In a restricted sense, the definition of Pharmacognosy implies a particular knowledge of methods of identification and evaluation of drugs.**

Definition of pharmacognosy

- Pharmacognosy may be defined as a branch of bioscience which treats in detail medicinal and related products of crude or primary type from plant, animal and mineral origins.
- **In short, it is an objective study of crude drugs from natural sources treated scientifically and it encompasses the knowledge of the history, distribution, cultivation, collection, processing for market, preservation, the study of sensory, physical, chemical and structural characters and the uses of crude drugs.**

- Pharmacognosy also includes study of other materials used in pharmacy such as suspending, disintegrating and flavouring agents, filtering aids, etc. And substances like antibiotics, allergens, hallucinogenic and poisonous plants, immunizing agents, pesticides, raw materials for the production of oral contraceptives etc.

History

- Nature always stands as a golden mark to exemplify the outstanding phenomenon of symbiosis.
- The plants are indispensable to man for his life.
- The three important necessities of life - food, clothing and shelter- and a host of other useful products are supplied to him by plant kingdom.
- Nature has provided a complete store-house of remedies to cure all ailments of mankind.
- The human being appears to be afflicted with more diseases than any other living species.
- In the past almost all the medicines used were from the plants, the plant being man’s only chemist for ages.
- Today a vast store of knowledge concerning therapeutic properties of different plants has accumulated.

History

- The history of herbal medicines is as old as human civilization.
- The documents, many of which are of great antiquity, revealed that plants were used medicinally in China, India, Egypt and Greece long before the beginning of the Christian era.
- One of the most famous surviving remnants is **Papyrus Ebers**, dating back to the sixteenth century before Christ. The text of document is dominated by more than 800 formulae and 700 different drugs.
- The drugs such as acacia, castor oil and fennel are mentioned along with apparent references to such compounds as iron oxide, sodium chloride, sodium carbonate and sulphur.

History

- Most of the medicinally active substances identified in the nineteenth and the twentieth centuries were used in the form of crude extract.
- Originally - during the 19th C and the beginning of the 20th C Pharmacognosy was used to define the branch of medicine or of commodity sciences, which dealt with **drugs in their crude, or unprepared form = crude drugs**

History

- In china, many medicinal plants had been in use since 5000 B.C.
- The oldest known herbal literature is Pen-t'sao written by emperor **Shen Nung** around 3000 B.C.
- It contains 365 drugs, one for each day of the year.
- Indians also worked meticulously to examine and classify the herbs which they came across, into groups called **Gunas**.
- **Charaka** made fifty groups of ten herbs each of which according to him, would suffice an ordinary physician's need. Similarly, **Sushrutha** arranged 760 herbs in 7 distinct wets based on some of their common properties.
- A large portion of Indian population even today depends on the Indian System of Medicine-**Ayurveda, 'An ancient science of life'.** The well known treatises in Ayurveda are **Charaka Samhita** and **Sushrutha Samhitha.**

History

- Following the contribution to medicine by **Hippocrates** (460-377 B.C) “Father of medicine”
- **Aristotle** (384-322 B.C)
- Greek Scientist, botonist and Philosopher **Theophrastus**(372-287B.C)
- **Dioscorides** (40-80 A.D)
- **Galen** (131-200A.D) and the early Arabian physicians there was a period of approximately 1000 years during which little, if any, progress was made in the medical sciences.
- The first pharmacist, **Galen**, was known to have had a number of pain-relieving materials, including opium in his apothecary.
- For the next few hundred years, the formulations of medicaments in use changed very little, although some bold attempts were made by chemical entrepreneurs such as **Paracelsus**(1493-1541) to develop mineral salts which might have had the potential of being universal curative agents.
- However, throughout this period, the use of herbal extracts for medicinal purposes never lost its place.

History

- The importance of extraction method and alcohol as an extractant was reported by **Le'mery** (1645-1715), **William withering** in 1785 published as account of some of the medicinal properties of foxglove leaves based on ten years of experimentation.
- Percolation process was used for the crude drugs.
- In 1803, the French Pharmacist, **Derosne** isolated narcotine from opium.
- In 1806, **Sertuerner** isolated morphine from opium and its role in alleviating pain was recognised.
- In the next few years, strychinine (1817), emetine (1817), brucine (1819), piperine (1819), quinine (1820) and colchicine (1820) were isolated.
- The French Pharmacist, **Pelletier**, first reported the isolation of strychnine from ignatius beans and latex of nuxvomica seeds.

History

- A new extraction process for alkaloids was developed by **Stass** and **Otto** in 1852.
- Other important discoveries during this period were the isolation of
  - nicotine from tobacco leaves(**Posselt** and **Reimann**,1828)
  - cocaine (**Neumann**, 1860)
  - ouabine (**Hardy** and **Gallows**, 1877)
  - pilocarpine (**Gerrard** and **Hardy**, 1875)
  - ephedrine(**Nagai**, 1887) and
  - podophyllotoxin (**kuersten**,1877).

- Thousands of plant and animal products used for the treatment of illness were described by Dioscorides in his book "**Dc Materia Medica**."
- Of these surprisingly a large number are still of importance in modern therapeutic practice: Aspidium, Cinnamon, Ergot, Hyoscyamus, and Opium were used in much the same manner as today.

History

- Isolation of ergometrine, digoxin, reserpine, theophylline and quinidine are the significant discoveries of the twentieth century.
- **Swede Linnaeus** (1707-1778) the great systematist classified the plants and introduced the system of naming the plants known as the binomial system which is still followed.

History

- The word "pharmacognosy" derives from the Greek words **pharmakon** (drug), and **gnosis** or knowledge. The term pharmacognosy was used for the first time by the Austrian physician Schmidt in 1811.
- Thus classically, Pharmacognosy was defined as the knowledge of drugs.
- In the nineteenth century, the term '**Materia Medica'** was used for the subject now known as 'Pharmacognosy'. While studying Sarsaparilla, it was **Seydler**, a German scientist, who coined the term 'Pharmacognosy' in 1815 in the title of his work "**Analecta Pharmacognostica**".
- Pharmacognosy is derived from two greek words, **Pharmakon** (a drug) and **Gignosco** (to acquire knowledge of).
- The study of **pharmakognosie** was first developed in German-speaking areas of Europe. The term drogenkunde ("science of crude drugs") is also used synonymously.

History

- **Materia Medica Meaning Medicinal Materials** From this humble beginning medicine and pharmacy gradually emerged along separate paths: the physician diagnosing the ailment and prescribing the remedy and the apothecary or pharmacist specializing in the collection, preparation, and compounding of the substance. Thus, the term materia medica (meaning medicinal materials) was synonymous with the substances and products derived from natural sources and employed by the physicians of that era.

History

- Originally - during the 19th century and the beginning of the 20th century - "Pharmacognosy" was used to define the branch of medicine or commodity sciences ("Warenkunde" in German), which dealt with drugs in their crude, or unprepared, form. Crude drugs are the dried, unprepared material of plant, animal or mineral origin, used for medicine.
- Modern Pharmacognosy focuses on the chemical components of the plant materials, including the structure and pharmacological properties that are responsible for their use in therapy.

Scope of Pharmacognosy

- The most comprehensive idea of the scope of Pharmacognosy was presented by Fluckiger, who stated that it "is the simultaneous application of various scientific disciplines with the object of acquiring knowledge of drugs from every point of view.“
- Pharmacognosy is one of the five major divisions of the pharmaceutical curriculum, represents the oldest branch of the profession of pharmacy.
- The ancients gathered herbs, animals, plants, and minerals and concocted them into ill-flavoured pungent mixtures.
- Innumerable remedies were known to the early practitioners of pharmacy and medicine, a fact indicated by the writings of Theophrastus, Pliny, Dioscorides, and their contemporaries.

Scope of Pharmacognosy

- *The scope of Pharmacognosy includes the study of the physical, chemical, biochemical and biological properties of drugs, drug substances, or potential drugs or drug substances of natural origin as well as the search for new drugs from natural sources.*
- Pharmacognosy as an applied science has played a crucial role in development of different disciplines of science.
- Pharmacognosy gives a sound knowledge of the vegetable drugs under botany and animal drugs under zoology.
- A Pharmacognosist should possess a sound knowledge of plant taxonomy, plant breeding, plant pathology and plant genetics, which is helpful in the development of cultivation technology for medicinal and aromatic plants.

- Traditionally the scope includes:
  - Identifying biological sources of drugs
  - Assessing the efficacy and safety of herbal drugs or their constituents as therapeutic agents using scientific methods
- Modern concept in Pharmacognosy
  - Securing regular supply of natural products

  ✓Cultivation

  ✓Increasing yield

  ✓Semi-synthesis

  ✓Synthesis
- Incorporation of a natural product in the system of modern medicine

- The paramount goal of pharmacognosy is
  - comprehensive investigations of plant materials by use of physical, chemical, and biological methods, and
  - also the search for a possibility to use these materials as natural medicines.
- Pharmacognosy is a broad discipline, which has adopted different concepts from various disciplines like chemistry, pharmacology, botany, agronomy, biotechnology, biochemistry and medicine.
- Other areas that have natural association with pharmacognosy are:
  - poisonous and hallucinogenic agents, allergens, herbicides, insecticides and molluscicides

- Although Pharmacognosy is principally concerned with medicinal plants,
  - There are a small number of animal products which are traditionally encompassed within it;
    - These include such items as beeswax, gelatin, wool fat, vitamins etc.
- Many of the animal kingdom, particularly marine organisms are receiving increasing attention
- Other natural products such as antibiotics, hormones and others may or may not be involved, depending on the teaching practice of a particular institution.
- Materials having no pharmacological action which are of interest to pharmacognosist are:
  - natural fibers, flavoring and suspending agents, colourants, disintegrants, stabilizers and filtering and support media

Scope of Pharmacognosy

- The knowledge of chemotaxonomy, biogenetic pathways for formation of medicinally active primary and secondary metabolites, plant tissue culture and other related fields is essential for complete understanding of Pharmacognosy.

A VITAL CONTRIBUTION TO THE ADVANCEMENT OF NATURAL AND PHYSICAL SCIENCE :

- This has done by the advanced technologies of cultivation, purification, identification (characterisation), of pharmaceuticals from nature.
- Concepts of biochemistry and chemical engineering help in the improvement of collection, processing and storage technologies of pharmaceuticals. It also gives knowledge of chemotaxonomy, biogenetic pathways for the formation of acute ingredients.

Scope of Pharmacognosy

A VITAL LINK BETWEEN PHARMACOLOGY AND MEDICINAL CHEMISTRY :

- Newly detected plant drugs are converting into medicine as purified phytochemicals.
- Pharmacognosy is essential for the evolution of new medicines because crude drugs are used for the preparation of galenicals or as a sources of therapeutically active metabolites.
- Phytopharmaceuticals that is synthesized drugs are finally formed in a suitable dosage forms and in which the crude drugs can be act as a intermediates.

Scope of Pharmacognosy

- In short Pharmacognosy is an important link between pharmaceuticals and basic science as well as ayurvedic and allopathic system of medicines.
- So Pharmacognosy is a science of active principles of crude drugs and which can be helpful in dispensing, formulating, manufacturing of dosage forms.
- In other way the complete knowledge of Pharmacognosy will help in the recent trend that is in industries.
- As a research tool and in new drug delivery systems, and all the departments of pharmaceuticals and one can improve the health care facilities across the world.
- Although most pharmacognostic studies focus on plants and medicines derived from plants, other types of organisms are also regarded as pharmacognostically interesting, in particular, various types of microbes (bacteria, fungi, etc.), and, recently, various marine organisms.

Scope of Pharmacognosy

- Pharmacognosy is interdisciplinary, drawing on a broad spectrum of biological and socio-scientific subjects, including **botany**, **ethnobotany**, **medical anthropology**, **marine biology**, **microbiology**, **herbalmedicine**, **chemistry**, **biotechnology**, **phytochemistry**, **pharmacology**, **pharmaceutics**, **clinical pharmacy** and **pharmacy practice**.
- The contemporary study of Pharmacognosy can be divided into the fields of
- **Ethnopharmacology**: the study of the pharmacological qualities of traditional medicinal substances; the study of **phytotherapy** (the medicinal use of plant extracts); and **phytochemistry**, the study of chemicals derived from plants (including the identification of new drug candidates derived from plant sources).

Scope of Pharmacognosy

- Medical ethnobotany: the study of the traditional use of plants for medicinal purposes
- **Zoopharmacognosy**, the process by which animals self-medicate, by selecting and using plants, soils, and insects to treat and prevent disease.
- Pharmcognosy-Biotechnology, the synthesis of natural bioactive molecules using biotechnology.
- Herbal interactions, the interactions of herbs with other drugs and body.
- **Marine pharmacognosy**, the study of chemicals derived from marine organisms.

Why do we study Pharmacognosy?

- Herbs and natural products offer a very good alternative source of drugs
  - The share of natural products in modern medicine is as high as 45 % in developed countries
  - In developing countries
    - ✓ About 80 % of the population, rely on natural remedies and
    - ✓ Almost every individual is believed to use herbal drugs at least occasionally.
  - The process is environmentally friendly.
  - Natural products are more tolerated by the body.
  - Plants are considered as big laboratories and mankind is said to have used less than 3-5 % of what nature can potentially provide.
- The study of natural products can generate valuable information for the synthetic chemist.